Which radar data format detects falls best?

Comparative Evaluation of mmWave Radar Data Representations for Contactless Fall Detection in Ambient Assisted Living

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Conclusions

- Point sets (Rep 3) detect falls best in unseen people: Macro F1 **83.0%**.
- They also give the fewest false alarms and are the fastest.
- Rep 1 stays near chance.

Impact: point-based processing suits privacy-preserving fall monitoring on low-cost radar.

Future work: larger groups and a threshold tuned on a validation set.

Background

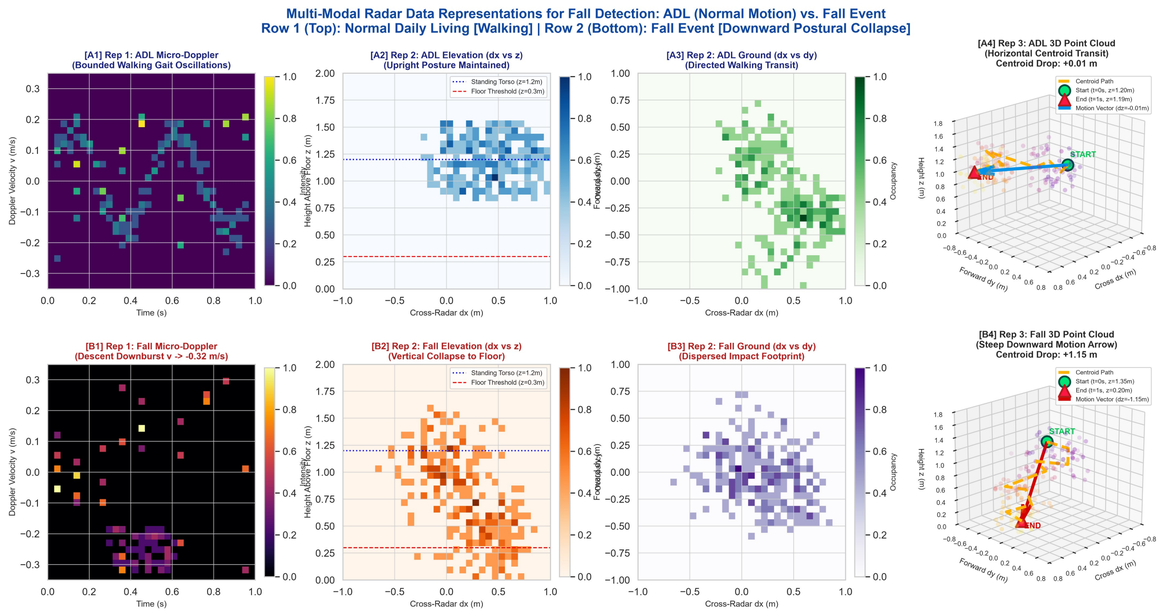
- Falls are a major cause of fatal injury among older adults.
- People do not wear devices, and cameras raise privacy concerns.

Why this research

- Radar protects privacy, but low-cost sensors give sparse point clouds.
- It is unclear which radar data format works best: earlier studies compare different networks or leak subject identity.

Objective: compare three radar representations on the same recordings, always testing on an unseen person.

Methods



Top row: ADL (walking).
Bottom row: Fall (downward collapse).

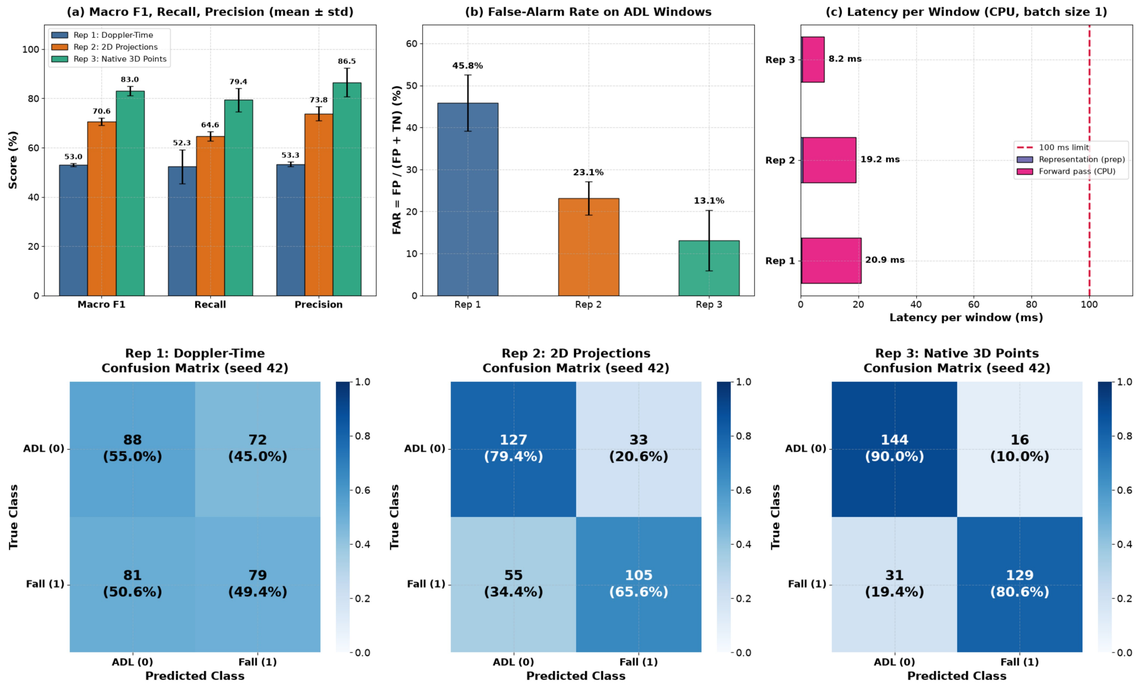
Rep 1: Doppler-time map
Velocity over time, 32 × 32. Fall: descent to −0.32 m/s.

Rep 2: 2D projections
Density maps XZ, XY and ZV. Fall: collapse to the floor.

Rep 3: 3D point set
5 features × 320 points. Fall: centroid drops 1.15 m, against 0.01 m when walking.

Pipeline: TI IWR6843 radar (3 subjects) → 10-frame windows of 32 points → ResNet-18 or PointNet-style → Fall or ADL, tested on unseen subjects with 3 seeds.

Results



	Rep 1	Rep 2	Rep 3
Macro F1	53.0 ± 0.6	70.6 ± 1.5	**83.0 ± 1.9**
ROC-AUC	55.2 ± 0.4	80.2 ± 1.2	**91.6 ± 2.3**
Recall	52.3 ± 7.0	64.6 ± 1.9	**79.4 ± 4.7**
False alarms	45.8 ± 6.7	23.1 ± 4.0	**13.1 ± 7.2**
Latency (ms)	20.9	19.2	**8.2**

Mean ± std over 3 seeds, in %.

Fig. 1: Rep 3 scores highest, raises the fewest false alarms and is fastest. All pipelines stay far below the 100 ms limit.

Fig. 3 (seed 42): Rep 3 raised 16 false alarms and missed 31 falls. Rep 1 raised 72 and missed 81.